

输入: 两个特征 $x_1, x_2 \in \{-1, +1\}$

输出: 感知器的输出 $y \in \{-1, +1\}$ 符号函数决定 $y = \text{sign}(w_1 x_1 + w_2 x_2 + b)$

画出真值表

· 列出 x_1, x_2 的所有组合

· 计算题目定义下的目标输出 y

INT301 Bio-computation

选择合适 w_1, w_2, b , 保证感知器输出匹配真值表.

判断是否线性可分:

· 如果正类与负类能用一条直线分开, 就能用感知器表示.

Week 2 Tutorial

Question 1 如果不分 (如 XOR 问题) 就不能用单层感知器表示.

The following two Boolean functions take two Boolean features x_1 and x_2 . The features can take on the values -1 and $+1$, where -1 represents False and $+1$ represents True. The output y of the functions can also take on the values -1 and $+1$, with the same interpretation. For each of the functions below, either give weights for a perceptron $y_0 = 1$

② x_1, x_2, y

x_1	x_2	y
+1	+1	-1
+1	-1	+1
-1	+1	-1
-1	-1	-1

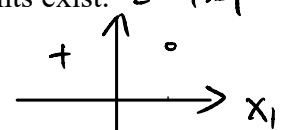
1) $y = \begin{cases} +1, & \text{if } x_1 = x_2 \\ -1, & \text{otherwise} \end{cases}$

2) $y = \begin{cases} +1, & \text{if } x_1 = 1 \text{ and } x_2 = -1 \\ -1, & \text{otherwise} \end{cases}$

可以线性可分
例如取: $w_1 = 1, w_2 = -1, b = -1$

① x_1, x_2, y

x_1	x_2	y
+1	+1	+1
+1	-1	-1
-1	+1	-1
-1	-1	+1



不可线性可分

Question 2 $(+1, -1): w_1 x_1 + w_2 x_2 + b =$

$$1(1) + (-1)(-1) - 1 = 1 > 0 \Rightarrow y = +1, \text{ 符合}$$

You want to design a neural network with sigmoid units to predict a person's academic role from his webpage. Possible roles are "professor", "student", and "staff". However, each person can take any number (from 0 to all 3) of these roles at the same time. Briefly,

describe: (1) 用一个三维向量表示角色标签, professor 代表第1位, student 第二位, staff 第三位

(2) Suggest a possible threshold value for the outputs. 1 表示有这个角色

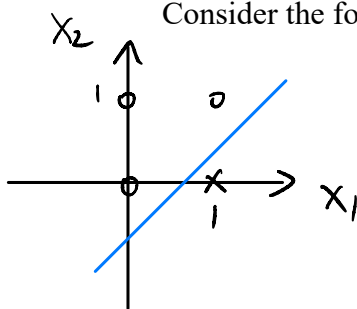
0 表示没有

(2) $\theta = 0.5$ 取中间值

只有教授 $[1, 0, 0]$
教授+学生 $[1, 1, 0]$
都不是 $[0, 0, 0]$

Question 3

Consider the following Boolean function:



A	B	$\neg A \vee B$
1	1	1
1	0	0
0	1	1
0	0	1

输出正类 1: $(A, B) = (1, 1), (0, 1), (0, 0)$

输出负类 0: $(A, B) = (1, 0)$

一个可能的感知器 w_1, w_2 与 b :

$$w_1 = -1, w_2 = 1, b = 0.5, b = w_0 x_0$$

$$(1, 1): -1 + 1 + 0.5 = 0.5 > 0 \quad x_0 = 1$$

$$(1, 0): -1 + 0 + 0.5 = -0.5 < 0$$

$$(0, 1): 0 + 1 + 0.5 = 1.5 > 0 \quad \theta = 0$$

$$(0, 0): 0 + 0 + 0.5 = 0.5 > 0 \quad w_0 = 0.5$$

Construct a perceptron that represents the function.

答案不唯一

Question 4

Consider a task involving "a three-input, one-output parity detector" which outputs a 1 if the number of "1" inputs is even; otherwise it outputs a 0. Can this function be represented by a perceptron? Explain.

把输入取为 $\{0, 1\}$ 三维向量, 偶校验 输出为 1 的正类: $P = \{000, 011, 101, 110\}$

负类: $N = \{001, 010, 100, 111\}$

若存在单层感知器 $w = (w_1, w_2, w_3)$, 阈值 b 使: $w^T x + b > 0 \quad (x \in P), \quad w^T x + b < 0 \quad (x \in N)$

正类

$$(0,0,0) \Rightarrow b > 0$$

$$(0,1,1) \Rightarrow w_2 + w_3 + b > 0$$

$$(1,0,1) \Rightarrow w_1 + w_3 + b > 0$$

$$(1,1,0) \Rightarrow w_1 + w_2 + b > 0$$

负类

$$(0,0,1) \Rightarrow w_3 + b < 0$$

$$(0,1,0) \Rightarrow w_2 + b < 0$$

$$(1,0,0) \Rightarrow w_1 + b < 0$$

$$(1,1,1) \Rightarrow w_1 + w_2 + w_3 + b < 0$$

化简: 正类 $2(w_1 + w_2 + w_3) + 3b > 0$ (A)

负类 $2(w_1 + w_2 + w_3) + 4b < 0$ (B)

但是 $(2(w_1 + w_2 + w_3) + 3b) + b$ 也应该 > 0 如果按A来说, 但这与B矛盾

所以 线性不可分.